Research Landscape of Time-Series Agent Systems

Model-side contributions, harness mechanisms, and evaluation resources

Model-Side Contributions

Representation · Alignment
Understanding · Adaptation

Representation & Alignment



- Time-LLM
- LLM4TS
- LLMTime
- ChatTS

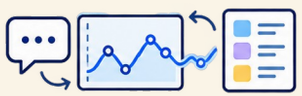
Learning & Reasoning



- Towards Time-Series Reasoning
- STRIDE
- OpenForecaster
- AURA-4B

Harness Mechanisms for Time-Series Agent Systems

1 Interface & Interaction



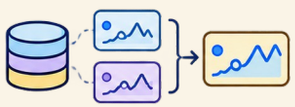
Structured Analysis Interfaces

- TS-Agent
- TimeART
- TSAG
- TESSA

Adaptive Interaction & Observation

- SAGE-Sleep
- ARTIST
- TimeRLM
- LLM-TSFD

2 Memory & Context



Retrieval & Context Construction

- Agentic RAG for TSA
- TimeCAP
- From News to Forecast
- ZARA
- MERIT

Temporal Experience & Evidence Memory

- FinMem
- MemCast
- ForecastCompass
- TIEM

3 Tools & Execution



Model & Code Execution

- TimeCopilot
- TimeSeriesScientist
- MoiraiAgent
- GenAutoML

Domain Tools & Simulation

- AgriWorld
- Open-TI
- GridMind
- PPG Physiological Agent

4 Control, Verification & Feedback



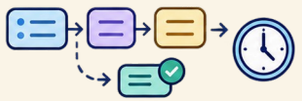
Review & Adjudication

- Argos
- AnomaMind
- TSQAgent
- Oil-Well Explainable Agent

Validation & Acceptance Gates

- Grid-Agent
- HydroAgent
- Neuro-Agentic Control
- BatteryLake

5 Planning & Workflow



Workflow Construction

- ClimateAgent
- DCATS
- MOSAIC
- Nexus

Operational Analysis Pipelines

- HFMD Structured Agents
- SIREN
- Smart Energy Agent
- S2S-FDD

6 Optimization & Evolution



Reflection & Experience Updates

- TimeClaw-EEL
- FLAIRR-TS
- CastFSR
- CRAFTER

Search & Adaptive Optimization

- MetaCaster
- EvoTS-Agent
- MarsTSC
- ATLAS

7 Coordination & Scaling



Debate & Evidence Aggregation

- TS-Debate
- MAS4TS
- Competition Agents
- FactorMAD

Coordinated Decision Systems

- FinCon
- TradingAgents
- CoLLMLight
- Fin-Analyst

Evaluation & Infrastructure

Benchmarks, datasets, simulators and task protocols for time-series agents

Benchmarks & Evaluation Resources



- TemporalBench
- TFRBench
- TraceBench
- TSAQA
- MMTS-Bench
- TimeSeriesGym
- TSQBench
- TimeSage-MT
- TS-Skill
- TimeSeriesExamAgent
- FutureSim
- WorldReasoner

Infrastructure & Task Protocols



- AION
- ModernTSF
- TimeClaw-Harness

Selected works grouped by relevant contributions. Subgroups provide descriptive reading aids; placements highlight selected mechanisms rather than exclusive paper categories.